QF620 Stochastic Modelling in Finance
Assignment 3/4
Due Date: 30-Oct-2024

1. Find the mean and variance of X_t if

   (a)
   $$dX_t = \mu dt + \sigma dW_t.$$

   (b)
   $$dX_t = \mu X_t dt + \sigma X_t dW_t.$$

   (c)
   $$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t.$$

2. Consider the two stochastic differential equations below:

   $$dX_t = rX_t dt + \sigma X_t dW_t$$
   $$dY_t = rY_t dt + \sigma Y_t d\tilde{W}_t$$

   Use Itô's formula to derive the stochastic differential equation for $Z_t = \frac{X_t}{Y_t}$, is

   (a) W_t and $\tilde{W}_t$ are independent.
   (b) W_t and $\tilde{W}_t$ have a correlation of 1.
   (c) W_t and $\tilde{W}_t$ have a correlation of ρ.

1. Find the mean and variance of X_t if

(a)

$$dX_t = \mu dt + \sigma dW_t.$$

intergating for both side: $W_t \sim N(0,t)$

$$\int_0^t dX_t = \int_0^t \mu dt + \int_0^t \sigma dW_t$$

$$X_t - X_0 = \mu(t-0) + \sigma(W_t - W_0)$$

$$X_t - X_0 = \mu t + \sigma W_t$$

$$X_t = X_0 + \mu t + \sigma W_t$$

$$E(X_t) = E(X_0 + \mu t + \sigma W_t)$$

$$= X_0 + \mu t$$

$$V(X_t) = V(X_0 + \mu t + \sigma W_t)$$

$$= \sigma^2 t$$

(b)

$$dX_t = \mu X_t dt + \sigma X_t dW_t.$$

let $Y_t = \log X_t = f(X_t)$

Itô's formula:

$$dY_t = f'(X_t) dX_t + \frac{1}{2} f''(X_t)(dX_t)^2$$

$$f'(X_t) = \frac{1}{X_t}$$

$$f''(X_t) = -\frac{1}{X_t^2}$$

$$(dX_t)^2 = (\mu dt + \sigma dW_t)^2$$

$$= \mu^2 (dt)^2 + 2\mu\sigma\, dt\, dW_t + \sigma^2 (dW_t)^2$$

$$= \sigma^2 dt$$

$$\therefore dY_t = \frac{1}{X_t} dX_t + \frac{1}{2}\left(-\frac{1}{\cancel{X_t^2}}\right) \cdot \sigma^2 \cancel{X_t^2} \cdot dt$$

$$= \frac{1}{X_t} \cdot (\mu \cancel{X_t} dt + \sigma \cancel{X_t} dW_t) - \frac{1}{2}\sigma^2 dt$$

$$dY_t = \mu dt + \sigma dW_t - \frac{1}{2}\sigma^2 dt$$

$$\int_0^t dY_t = \int_0^t (\mu - \frac{1}{2}\sigma^2) dt + \int_0^t \sigma dW_t$$

$$Y_t - Y_0 = (\mu - \frac{1}{2}\sigma^2)(t-0) + \sigma(W_t - W_0)$$

$$\log\left(\frac{X_t}{X_0}\right) = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

$$\log X_t = \log X_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

$$X_t = X_0 \cdot e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

$$E(X_t) = X_0 \cdot E\left(e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}\right)$$

$$= X_0 \cdot e^{(\mu - \frac{1}{2}\sigma^2)t} \cdot E(e^{\sigma W_t})$$

$\because W_t \sim N(0, t)$

$\sigma W_t \sim N(0, \sigma^2 t)$

if $Z \sim N(0, \sigma^2 t)$, $E(e^Z) = e^{\frac{1}{2}Var(Z)}$

So let $\sigma W_t = Z$

$$\therefore E(X_t) = X_0 \cdot e^{(\mu - \frac{1}{2}\sigma^2)t} \cdot e^{\frac{1}{2}\sigma^2 t}$$

$$= X_0 \cdot e^{\mu t}$$

$$X_t^2 = X_0^2 \cdot e^{2(\mu - \frac{1}{2}\sigma^2)t + 2\sigma W_t}$$

$$E(X_t^2) = E\left[X_0^2 \cdot e^{2(\mu - \frac{1}{2}\sigma^2)t + 2\sigma W_t}\right]$$

$$= X_0^2 \cdot e^{2(\mu - \frac{1}{2}\sigma^2)t} \cdot E(e^{2\sigma W_t})$$

$\because W_t \sim N(0, t)$

$\therefore 2\sigma W_t \sim N(0, 4\sigma^2 t)$

$\because Z \sim N(0, \sigma^2)$, $E(e^Z) = e^{\frac{1}{2}V(Z)}$

$\therefore$ let $Z = 2\sigma W_t$

$$\therefore E(X_t^2) = X_0^2 \cdot e^{2(\mu - \frac{1}{2}\sigma^2)t} \cdot e^{\frac{1}{2}(4\sigma^2 t)}$$

$$= X_0^2 \cdot e^{2(\mu - \frac{1}{2}\sigma^2)t} \cdot e^{2\sigma^2 t}$$

$$= X_0^2 e^{2\mu t-\sigma^2 t} \cdot e^{2\sigma^2 t}$$

$$E(X_t^2) = X_0^2 \cdot e^{2\mu t+\sigma^2 t}$$

$$V(X_t) = E(X_t^2) - E(X_t)^2$$

$$= X_0^2 \cdot e^{2\mu t+\sigma^2 t} - (X_0 e^{\mu t})^2$$

$$= X_0^2 \cdot e^{2\mu t+\sigma^2 t} - X_0^2 \cdot e^{2\mu t}$$

$$= X_0^2 (e^{2\mu t+\sigma^2 t} - e^{2\mu t})$$

$$= X_0^2 \cdot e^{2\mu t}(e^{\sigma^2 t} - 1)$$

(c)

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t.$$

applying Itô's formula: $X_t = e^{kt} X_t = f(t, X_t)$

$$df(e^{kt} X_t) = f'(t)dt + f'(X_t)dX_t + \tfrac{1}{2} f''(X_t)(dX_t)^2$$

$$f'(t) = ke^{kt} \cdot X_t$$

$$f'(X_t) = e^{kt}$$

$$f''(X_t) = 0$$

$$df(X_t) = k \cdot e^{kt} X_t dt + e^{kt} dX_t + \tfrac{1}{2} \cdot 0 (dX_t)^2$$

$$= ke^{kt} X_t dt + e^{kt} dX_t$$

$$= ke^{kt} X_t dt + e^{kt} \cdot [k(\theta - X_t)dt + \sigma dW_t]$$

$$= ke^{kt} \cdot X_t dt + e^{kt}(k\theta - kX_t dt + \sigma dW_t)$$

$$= \cancel{ke^{kt} \cdot X_t dt} + e^{kt} \cdot k\theta - \cancel{e^{kt} \cdot kX_t dt} + e^{kt} \cdot \sigma dW_t$$

$$= e^{kt} \cdot k\theta \, dt + e^{kt} \cdot \sigma dW_t$$

$$\int_0^t df(X_s) = \int_0^t e^{ks} \cdot k\theta ds + \int_0^t e^{ks} \cdot \sigma dW_s$$

$$f(X_t) - f(X_0) = [\theta e^{ks}]_0^t + \sigma \int_0^t e^{ks} \cdot dW_s$$

$$\because f(X_t) = e^{kt} X_t$$

$$\therefore f(X_0) = e^0 X_0$$

$$e^{kt} X_t - e^0 X_0 = \theta e^{kt} - \theta e^0 + \sigma \int_0^t e^{ks} \cdot dW_s$$

$$X_t = X_0 \cdot e^{-kt} + \theta(1-e^{-kt}) + \sigma\int_0^t e^{k(w-t)} dW_s$$

$$E(X_t) = E[X_0 \cdot e^{-kt} + \theta(1-e^{-kt})] + E[\sigma\int_0^t e^{k(w-t)} dW_s]$$

$\because E(dW_s) = 0$

$$E(X_t) = X_0 e^{-kt} + \theta(1-e^{-kt})$$

from X_t, we can caculate X_t^2:

$$X_t^2 = \left[X_0 e^{-kt} + \theta(1-e^{-kt}) + \sigma\int_0^t e^{k(w-t)} dW_s\right]^2$$

$$= (X_0 e^{-kt} + \theta(1-e^{-kt}))^2 + 2(X_0 e^{-kt} + \theta(1-e^{-kt}))\cdot\sigma\int_0^t e^{k(w-t)} dW_s + \sigma^2\left(\int_0^t e^{-k(t-s)} dW_s\right)^2.$$

$$\therefore E(X_t^2) = (X_0 e^{-kt} + \theta(1-e^{-kt}))^2 + 0 + E(\sigma^2\int_0^t e^{-k(t-s)} dW_s)^2)$$

$$= (X_0 e^{-kt} + \theta(1-e^{-kt}))^2 + \sigma^2\int_0^t e^{-2k(t-s)} ds$$

$$= (X_0 e^{-kt} + \theta(1-e^{-kt}))^2 + \frac{\sigma^2}{2k}(1-e^{-2kt})$$

$$V(X_t) = E(X_t^2) - E(X_t)^2$$

$$= (X_0 e^{-kt} + \theta(1-e^{-kt}))^2 + \frac{\sigma^2}{2k}(1-e^{-2kt}) - [X_0 e^{-kt} + \theta(1-e^{-kt})]^2$$

$$= \frac{\sigma^2}{2k}(1-e^{-2kt})$$

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Use Itô's formula to derive the stochastic differential equation for $Z_t = \frac{X_t}{Y_t}$, is

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(c) W_t and $\tilde{W}_t$ have a correlation of ρ.

$$f(X_t, Y_t) = Z_t = \frac{X_t}{Y_t}$$

$$\frac{\partial f}{\partial X_t} = \frac{1}{Y_t}$$

$$\frac{\partial^2 f}{\partial X_t^2} = 0$$

$$\frac{\partial f}{\partial Y_t} = -\frac{X_t}{Y_t^2}$$

$$\frac{\partial^2 f}{\partial Y_t^2} = \frac{2X_t}{Y_t^3}$$

$$\frac{\partial^2 f}{\partial X_t Y_t} = -\frac{1}{Y_t^2}$$

$$dZ_t = \frac{\partial f}{\partial X_t}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial X_t^2}(dX_t)^2 + \frac{\partial f}{\partial Y_t}dY_t + \frac{1}{2}\frac{\partial^2 f}{\partial Y_t^2}(dY_t)^2 + \frac{\partial^2 f}{\partial X_t Y_t}dX_t dY_t.$$

$$= \frac{1}{Y_t}dX_t + 0 - \frac{X_t}{Y_t^2}dY_t + \frac{1}{2}\cdot\frac{2X_t}{Y_t^3}(dY_t)^2 + -\frac{1}{Y_t^2}dX_t dY_t$$

① $\frac{1}{Y_t}dX_t = \frac{1}{Y_t}\cdot(rX_t dt + \sigma X_t dW_t)$

$$= r\frac{X_t}{Y_t}dt + \sigma\frac{X_t}{Y_t}dW_t$$

$$= r\cdot \cancel{Z_t dt} + \sigma Z_t dW_t$$

② $-\frac{X_t}{Y_t^2}dY_t = -\frac{X_t}{Y_t^2}(r\cancel{Y_t}dt + \sigma \cancel{Y_t}d\hat{W}_t)$

$$= -r\frac{X_t}{Y_t}dt - \sigma\frac{X_t}{Y_t}d\hat{W}_t$$

$$= -r\cdot\cancel{Z_t dt} - \sigma Z_t d\hat{W}_t$$

③ $\frac{1}{2}\cdot\frac{2X_t}{Y_t^3}(dY_t)^2 = \frac{1}{\cancel{2}}\frac{\cancel{2}X_t}{Y_t^3}[(rY_t dt)^2 + 2(rY_t dt)(\sigma Y_t d\hat{W}_t) + (\sigma Y_t d\hat{W}_t)^2]$

$$= \frac{X_t}{\cancel{Y_t^3}}(\sigma^2 Y_t^2 d\hat{W}_t^2)$$

$$= Z_t\sigma^2 dt$$

④ $-\frac{1}{Y_t^2}dX_t dY_t = -\frac{1}{Y_t^2}\sigma^2 X_t Y_t \rho\, dt$

$$\therefore dZ_t = \sigma Z_t dW_t - \sigma Z_t d\hat{W}_t + Z_t\cdot\sigma^2 dt - \frac{1}{Y_t^2}\sigma^2 X_t Y_t \rho\, dt$$

$$= \sigma Z_t(dW_t - d\hat{W}_t) + Z_t\cdot\sigma^2 dt - Z_t\sigma^2\rho\, dt$$

$$= \sigma Z_t(dW_t - d\hat{W}_t) + Z_t\sigma^2 dt(1-\rho)$$

ⓐ if W_t and $\hat{W}_t$ are independant

$\therefore \rho = 0.$

$\therefore dZ_t = \sigma Z_t(dW_t - d\hat{W}_t) + Z_t\cdot\sigma^2 dt$

ⓑ $\rho = 1$, W_t and $\hat{W}_t$ can be a same brownian motion.

$\therefore dW_t - d\hat{W}_t = 0$

$\therefore dZ_t = \sigma Z_t(dW_t - d\hat{W}_t) = 0$

ⓒ $\rho = \rho$

$dZ_t = \sigma Z_t(dW_t - d\hat{W}_t) + Z_t\sigma^2 dt(1-\rho)$